Joshua Denbow | CS499 | Milestone 2 Narrative | 9/26/2026

The artifact I selected for this enhancement is my *InventoryApp*, an Android application I originally created in CS 360: Mobile Architect and Programming. The original application allowed users to create an account, log in, and manage inventory through basic add, update, and delete functions. It also included an SMS notification for items that reached zero stock. While the application worked, much of the UI logic, business logic, and database interaction was handled directly through the Activities and *DatabaseHelper* class, leaving plenty of room for improvement.

I selected this artifact because it gave me a good opportunity to demonstrate how my software engineering skills have developed. I reorganized the application into a more modular structure by adding an *InventoryItem* model, *InventoryRepository* and *UserRepository* classes for data access, and an *InventoryService* for validation and business logic. I also replaced the original inventory display with a *RecyclerView* and reusable adapter. The updated dashboard now provides inventory totals along with searching, filtering, and sorting. I also added a separate item detail screen for editing and deleting items, improved input validation and user feedback, and added confirmation before deletion.

Security was another area I improved. Instead of storing passwords directly as plaintext, the enhanced application uses salted PBKDF2 password hashing through a separate *PasswordUtils* class. Authentication and database access were also moved out of the Activities, and parameterized database selections are used where appropriate. These changes demonstrate a stronger security mindset while also improving the organization and maintainability of the application.

This enhancement as described above and shown in the files I submitted supports the course outcomes by demonstrating stronger software design, appropriate use of computing techniques, clearer communication of information through the improved interface, and consideration of security during development. One of the primary challenges for me was restructuring an application that already worked without breaking its existing functionality (an issue I have on occasion). I handled this by making the changes in separate stages and repeatedly compiling and testing account creation, authentication, inventory CRUD operations, validation, searching, filtering, sorting, and data persistence. Overall, the enhancement gave the original application a much cleaner structure and a stronger foundation for the algorithm and database enhancements I will complete later in the capstone.